

Two Incomparable Denominator Exclusions for $\sum_{n \geq 2} (n! - 1)^{-1}$

Finite certificates and exact rationality criteria for Erdős Problem #68

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ABSTRACT

Every rational representation $S = a/q$, $q > 0$, of $S = \sum_{n \geq 2} (n! - 1)^{-1}$ satisfies

$$q \nmid 299999!, \quad q \geq 2^{39990} > 10^{12038}.$$

The first exclusion follows from an exact strict-successor carry test; the second from continued fractions. An integral divisor basis determines the cancelling vectors and their attainable factorial moments. At fixed moment, every correction translates the full residual by an integer. Actual prime-power cancellations explain why common-denominator growth alone does not control reduced prefixes. The companion constant $S - e + 2$ gives an exact factorial-orbit criterion; the remaining assertion is strict-successor divisibility failure at arbitrarily large indices.

1 The denominator exclusions

Let $S = \sum_{n \geq 2} (n! - 1)^{-1}$ and

$$H_m = \sum_{n=2}^m \frac{1}{n! - 1}, \quad Z_m = \lfloor m!H_m \rfloor + 1, \quad b_m = mZ_{m-1} + 1 - Z_m.$$

The divisibility exclusion in the abstract rests on the implication

$$S = a/q, \quad q \mid (m-1)!, \quad m \geq 3 \implies b_m = 1. \quad (1)$$

Indeed, $0 < m!(S - H_m) < 1$, by comparison with the telescoping series $\sum_{n > m} (n-1)/n! = 1/m!$. Thus $m!S$ is the unique integer strictly above $m!H_m$. The same argument at $m-1$ gives $Z_m = m!S = mZ_{m-1}$, proving (1). The exact value $b_{300000} \neq 1$ therefore excludes every divisor of 299999! as a denominator of S .

The independent continued-fraction exclusion bounds the denominator's size. Neither restriction implies the other: a prime between 299999 and 599998 satisfies the divisibility restriction and fails the size restriction, whereas 299999! does the reverse. Both

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conclusions are finite; they do not decide the irrationality question posed by Erdős [1, 2]. Their computational certificates are specified in §8.

Adjacent factorial differences affect exactly their divisor channels. Integral triangular elimination therefore solves the channel equations. The support restriction determines which factorial moments occur, while at a fixed moment every correction changes the residual by an integer. The argument separates coefficient feasibility from the real separation needed for irrationality.

2 The fixed companion orbit

The termwise identity

$$\frac{1}{n! - 1} = \frac{1}{n!} + \frac{1}{n!(n! - 1)}$$

replaces the changing prefixes by the factorial orbit of one fixed number,

$$C = \sum_{n \geq 2} \frac{1}{n!(n! - 1)}.$$

Absolute convergence gives

$$C + (e - 2) = S. \quad (2)$$

The endpoint of the exponential prefix contributes residue 1 modulo m ; its positive tail removes another unit on taking the floor. This accounts for the distinguished residue -2 in the following theorem.

Theorem 2.1 (fixed companion-orbit rationality boundary). *The following statements are equivalent:*

1. $S \in \mathbb{Q}$;
2. $(\lfloor m!C \rfloor + 2) \bmod m = 0$ for every sufficiently large m .

Consequently,

$$S \notin \mathbb{Q} \iff (\forall B)(\exists m > B) (\lfloor m!C \rfloor + 2) \bmod m \neq 0. \quad (3)$$

The equivalence and its cofinal form are [Lean source](#).

Proof. First suppose $S = a/q$ with $q > 0$, and take m so large that $q \mid (m - 1)!$. Write

$$E_m = m! \sum_{2 \leq n \leq m} \frac{1}{n!}, \quad \varepsilon_m = m! \sum_{n > m} \frac{1}{n!}.$$

Then E_m is an integer, $0 < \varepsilon_m < 1$, and $m!S$ is an integer divisible by m . Multiplying (2) by $m!$ therefore gives

$$\lfloor m!C \rfloor = m!S - E_m - 1.$$

Every summand of E_m except the endpoint $m!/m! = 1$ is divisible by m . Thus $E_m \equiv 1 \pmod{m}$ and $\lfloor m!C \rfloor \equiv -2 \pmod{m}$.

Conversely, assume the displayed congruence from some index onward. Let $d_m(C) = \lfloor m!C \rfloor - m\lfloor (m-1)!C \rfloor$ be the canonical factorial digit. Since $0 \leq d_m(C) < m$, for $m \geq 3$ the congruence is equivalent to

$$d_m(C) = m - 2.$$

Choose N beyond the exceptional indices. The canonical factorial expansion of C then has the form

$$C = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C)}{m!} + \sum_{m>N} \frac{m-2}{m!}.$$

Adding $e - 2 = \sum_{m \geq 2} 1/m!$ and using the telescoping identity

$$\sum_{m>N} \frac{m-1}{m!} = \sum_{m>N} \left(\frac{1}{(m-1)!} - \frac{1}{m!} \right) = \frac{1}{N!}$$

gives

$$S = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C) + 1}{m!} + \frac{1}{N!},$$

which is rational. Negating the eventual statement yields the cofinal formulation above. $\square$

Remark. The remaining assertion is cofinal escape of this actual factorial orbit from -2 . The theorem identifies the required event without supplying it.

2.1 The carry and its wrap correction

The canonical factorial digit is $d_m(C) = \lfloor m!C \rfloor - m\lfloor (m-1)!C \rfloor$. For $C_m = \sum_{n=2}^m 1/(n!(n!-1))$, put

$$\delta_m = m!(C - C_m), \quad \sigma_m = \mathbf{1}_{\{\lfloor m!C \rfloor < \delta_m\}}.$$

The positive tail satisfies $0 < \delta_m < 1/((m+1)! - 1)$. Subtracting it crosses an integer exactly when $\sigma_m = 1$, so for $m \geq 3$

$$b_m = m - 1 - d_m(C) + \sigma_m - m\sigma_{m-1}. \quad (4)$$

This identity is unconditional. A floor-stability hypothesis is required only to discard the two wrap terms. Equality in the defining comparison is the no-wrap case. Canonical factorial expansions and their rationality criterion are classical [3, 4].

Theorem 2.2 (exact carry characterisation). *The following conditions are equivalent:*

$$S \notin \mathbb{Q}, \quad (\forall B)(\exists m > B) b_m \neq 1, \quad (\forall B)(\exists m > B) m \nmid Z_m.$$

The pair of equivalences is [Lean source](#).

Proof. Rationality forces eventual unit carries by (1). Conversely, if $b_m = 1$ eventually, then $Z_m/m!$ is eventually constant. Since

$$H_m < Z_m/m! \leq H_m + 1/m!,$$

that rational constant is S . Finally writing $x = (m-1)!H_{m-1}$ gives $b_m = m - 1 - \lfloor m\{x\} + 1/(m! - 1) \rfloor$, hence $-1 \leq b_m \leq m - 1$. For $m \geq 3$, these bounds together with $Z_m = mZ_{m-1} + 1 - b_m$, imply $m \mid Z_m$ exactly when $b_m = 1$. $\square$

3 An integral basis for factorial channels

For a finite integer vector λ , define

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}, \quad M(\lambda) = \sum_n \lambda_n n!, \quad V_d(\lambda) = \sum_n \lambda_n W_{d,n}.$$

The weights are integers. We first allow the auxiliary coordinate e_1 , then impose the manuscript support $n \geq 2$. This coefficient convention introduces no term $1/(1! - 1)$ into S .

Theorem 3.1 (divisor-channel coordinates). *Set*

$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \quad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \dots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \quad (5)$$

The observables, the unique expansion and the coefficient formula are [Lean source](#).

Proof. The moment of T_n is zero. The floor in $W_{d,n}$ changes between $n-1$ and n precisely when $d \mid n$, so

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d|n}.$$

The recursion cancels the proper divisor channels, proving the two identities by induction. The vectors $e_1, T_2, \dots, T_N$ form an integral triangular basis with final coefficients $1, -1, \dots, -1$. The change from T_n to U_n is integral triangular with unit diagonal. Hence $e_1, U_2, \dots, U_N$ is also an integral basis. Applying M and each V_d identifies its coefficients as in (5). For d beyond the support, $V_d = M$, so no infinite sum is required. $\square$

The prime translator is the case $U_p = T_p$ for a prime $p \geq 3$. The recursion also isolates composite channels; for example

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \quad (6)$$

It has moment zero and only channel 9 is nonzero. Thus each channel can be adjusted separately by a multiple of its own denominator. The expansion is [Lean source](#) and the observable profile is [Lean source](#).

3.1 The support restriction and the residual

Write

$$L_D = \text{lcm}_{2 \leq d \leq D}(d! - 1), \quad K_D = L_D e_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

Equation (5) implies both the channel congruence

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1} \quad (7)$$

in the equivalent integer-divisibility form [Lean source](#), and the unique description of every low-channel kernel:

$$V_2 = \dots = V_D = 0 \iff \lambda = tK_D + \sum_{n>D} z_n U_n, \quad M = tL_D. \quad (8)$$

For the full residual

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1},$$

we have $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$. Therefore

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n. \quad (9)$$

The series converges because $V_d = M$ beyond the support. This equation identifies the limit of channel adjustment: every zero-moment correction changes the residual by an integer. A further fractional cancellation coordinate does not arise from these corrections.

Only finitely many z_n are nonzero. Put $a_D = [e_1]K_D$ and $u_n = [e_1]U_n$. Admissibility on $n \geq 2$ becomes the single integer equation

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (10)$$

Consequently the attainable moments are

$$L_D \frac{g_D}{\gcd(g_D, a_D)} \mathbb{Z}, \quad g_D = \gcd\{u_n : n > D\}. \quad (11)$$

Indeed, finite integer combinations of the u_n form $g_D \mathbb{Z}$. The resulting positive generator is attained, and an attaining vector is primitive. Odd u_n vanish by induction. For a prime p , the sole nonzero proper-divisor contribution at $2p$ gives $u_{2p} = -2(2p)!/2^p \neq 0$. Every tail thus contains a nonzero coefficient, so $g_D > 0$.

A finite certificate for the infinite moment ideal. The scalar recurrence is

$$u_2 = 2, \quad u_n = - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

Once a divisor g controls the computed tail, only the terms with $d \leq D$ can supply a nonzero residue to a later coefficient. Those finitely many sources have a factorial cutoff:

$$k! \mid W_{d,dk} = \frac{(dk)!}{(d!)^k}.$$

The quotient by $k!$ counts partitions into k unordered blocks of size d .

Theorem 3.2 (finite determination of the moment ideal). *Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then*

$$g_D = \gcd(u_{D+1}, \dots, u_H), \quad H < 2D^2.$$

The statement is [Lean source](#).

Proof. The finite gcd g divides the nonzero term $u_{2\ell} = -(2\ell)!/2^{\ell-1}$, so $g \mid (2\ell)!$. For $n > H$ and a divisor $d \leq D$, the integer n/d is at least 2ℓ . Hence $g \mid (n/d)! \mid W_{d,n}$. Every early source is divisible by g . Strong induction handles the remaining divisors $D < d < n$, whose coefficients are already divisible by g . Thus g divides the entire tail. Bertrand's postulate supplies ℓ for $D \geq 3$; take $\ell = 2$ for $D = 2$. Finally $H \leq D(2D - 1) < 2D^2$. $\square$

The earlier envelope $\Lambda_n = \text{lcm}(1, \dots, n) \mid u_n$ remains useful for other certificates. Finite-source checks can end sooner because only the sum of the sources has to vanish modulo g . The even-coefficient sign pattern supplies a nonzero term in every tail. Together with (11) this determines the unrestricted minimum positive moment for every fixed depth. For example, $L_4 = 115$, $a_4 = -55$, $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60 \mid \Lambda_9$ give minimum moment 1380, attained by $12K_4 + 253U_6 - 11U_8$. On support through 6 the minimum is instead 4140. Minimum moment and minimum upper support are different objectives. Neither computation supplies a cofinal real-residual gap, and neither alters the two finite denominator exclusions.

4 The moment cost and the choice of support

On the actual support $n \geq 2$, one has the stronger congruence $12 \mid M - 2V_2$. For $n = 2, 3$, $n! = 2W_{2,n}$. For $n \geq 4$, both $n!$ and $2W_{2,n}$ are divisible by 12: writing $n = 2k$ or $2k + 1$, $W_{2,2k} = k! \prod_{j=1}^k (2j - 1)$ is divisible by 6 for $k \geq 2$. Since every $d! - 1$ is coprime to 6, low-channel annihilation implies

$$12L_D \mid M. \tag{12}$$

The factor 12 is sharp at $D = 2$ and $D = 3$, witnessed respectively by $-6e_2 + e_4$ and $-6e_2 - 8e_3 + 5e_4$.

4.1 Growth of the compulsory factor

The following elementary estimate quantifies this cost:

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}. \tag{13}$$

The extended-real statement, with the real exponent $3/2$, is [Lean source](#). For a terminal block of k terms, the product–lcm–gcd inequality and $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$ give

$$\log L_N \geq \sum_{n=N-k+1}^N \log(n! - 1) - \binom{k+1}{3} \log N.$$

Taking $k = \lfloor \alpha \sqrt{N} \rfloor$ gives leading coefficient $\alpha - \alpha^3/6$, maximised by $\alpha = \sqrt{2}$. The same argument applies to every fixed nonzero polynomial perturbation $n! + P(n)$, using

$$\gcd(i! + P(i), j! + P(j)) \mid P(j) - (j!/i!)P(i);$$

the additional $O_P(k^2 \log N)$ loss is lower order. The subtraction identity is recorded by Lai [6, (2.5)]. For $P = 0$, the lcm is only $N!$. These are common-denominator estimates. In particular, $L_N(S - H_N) \rightarrow \infty$, so multiplying the prefix by its full lcm cannot make its positive tail small.

4.2 Remote primitive kernels

Support location imposes a different requirement from low moment. Let $D, r \geq 2$ and let L be a positive multiple of $2, \dots, D$. Put

$$i_j = r + jL \ (0 \leq j < D), \quad N = r + (D-1)L, \quad \alpha_d = (d!)^{L/d},$$

$$H(X) = \prod_{d=2}^D (\alpha_d X - 1) = \sum_{j=0}^{D-1} h_j X^j, \quad A = \prod_{d=2}^D \alpha_d.$$

The primitive kernel on this grid is

$$\lambda_j^* = \frac{N! h_j}{A i_j!}, \quad M = N! \prod_{d=2}^D (1 - 1/\alpha_d) > 0.$$

Its final coefficient is 1. The channel equations are polynomial vanishing at the distinct points α_d^{-1} . Integrality follows by expanding h_j/A into products of reciprocal α_d : each product divides the factorial quotient $N!/i_j!$. The minimum step is $L = \text{lcm}(2, \dots, D)$.

The stronger divisibility

$$r! \prod_{d=2}^D (L/d)! \prod_{d=2}^D (\alpha_d - 1) \mid M$$

follows by counting partitions into the specified equal-sized blocks. In particular $\lfloor \sqrt{N}/2 \rfloor! \mid M$: if $v = \max(r, L/2)$, then $N < 4v^2$ and $v! \mid M$. Thus these moments absorb every fixed denominator as $N \rightarrow \infty$. Cofinal residual nonintegrality on this family is sufficient for irrationality; the construction does not establish that nonintegrality.

5 Cancellation in a reduced prefix

A common-denominator bound does not determine the denominator after addition. Put $L_M = \text{lcm}_{2 \leq n \leq M}(n! - 1)$ and $A_M = \sum_{n=2}^M L_M / (n! - 1)$. For a prime q with attained maximum $e = \max_{2 \leq n \leq M} v_q(n! - 1) > 0$, set

$$I_{q,e}(M) = \{n \in [2, M] : v_q(n! - 1) = e\}, \quad R_{q,e}(M) = \sum_{n \in I_{q,e}(M)} \left(\frac{n! - 1}{q^e} \right)^{-1} \pmod{q}.$$

After multiplication by L_M , all lower-valuation terms vanish modulo q . Hence

$$A_M \equiv (L_M / q^e) R_{q,e}(M) \pmod{q},$$

and L_M / q^e is a unit. Consequently

$$v_q(\text{den } H_M) = e \iff R_{q,e}(M) \not\equiv 0 \pmod{q}. \quad (14)$$

The equivalence, with the maximum and its attainment taken from the actual prefix lcm, is [Lean source](#). This first-layer test is a consequence of the general reciprocal-sum valuation formula of Louwsma and Martino [5, Lemma 4.1].

The actual factorial-gap examples show why the weights matter. At $M = 138$, the maximal 139-hits are 69, 122, 137, with cofactor residues 6, 49, 73, and

$$6^{-1} + 49^{-1} + 73^{-1} \equiv 0 \pmod{139}.$$

At $M = 2592$, the maximal 2593-hits are 349, 2243, 2591, with residues 1508, 1566, 1678 and reciprocal sum zero. All these attained exponents are one, so the respective prime disappears completely from the reduced prefix. Singleton maximal support cannot cancel, whereas a hit-count bound alone cannot determine a weighted sum. Even survival must subsequently be coupled to a real strict-successor inequality.

6 One joint collision and residue estimate

The whole-prefix construction separates the common collision factor from prime-power layers with a unique maximal owner. For a natural parameter $p \geq 3$, write $d_n = n! - 1$ and put

$$\begin{aligned} F_p &= (p-1)!, \quad D_p = \text{lcm}_{2 \leq i < j \leq 2p-1} \gcd(d_i, d_j), \\ C_p &= \text{lcm}(F_p, D_p), \quad L_p^{\text{blk}} = \text{lcm}(F_p, d_2, \dots, d_{2p-1}), \\ R_p &= L_p^{\text{blk}} / C_p, \quad T_p = \sum_{n=2}^{2p-1} L_p^{\text{blk}} / d_n, \quad \rho_p = (-T_p) \bmod R_p. \end{aligned}$$

Every prime dividing R_p has a unique denominator attaining its maximal valuation. Reducing T_p modulo that prime leaves one unit term, giving $\gcd(T_p, R_p) = 1$. Thus $0 < \rho_p < R_p$ whenever $R_p > 1$.

Proposition 6.1 (global complementary-residue criterion). *If arbitrarily large natural parameters $p \geq 3$ satisfy*

$$R_p > 1, \quad (2p+1)L_p^{\text{blk}} < 2p^2(2p-1)!\rho_p, \quad (15)$$

then S is irrational.

The criterion at natural parameters is [Lean source](#).

Proof. Suppose $S = a/q$. For large p , $q \mid F_p \mid C_p$, so $L_p^{\text{blk}}S$ is an integer multiple of R_p . Therefore

$$A = L_p^{\text{blk}}(S - H_{2p-1})$$

is a positive integer congruent to $-T_p$ modulo R_p , and $A \geq \rho_p$. The positive tail bound

$$S - H_{2p-1} < \frac{2p+1}{2p^2(2p-1)!}$$

and (15) give $A < \rho_p$, a contradiction. $\square$

Let $\tilde{C}_p = C_p/F_p$ and $\eta_p = \rho_p/R_p$. The inequality is exactly

$$\log \tilde{C}_p - \log \eta_p < \log \frac{2p^2(2p-1)!}{(2p+1)(p-1)!}. \quad (16)$$

The right side is $p \log p + (2 \log 2 - 1)p + O(\log p)$. The modulus R_p cancels: increasing private support alone does not improve this comparison. Coprimality permits $\rho_p = 1$, so the complementary gap needs its own quantitative control. Both losses must be controlled at the same parameters. A joint mean bound below $(1-\varepsilon)p \log p$ on the same nonempty sets in $[X, 2X]$ would suffice; two unrelated cofinal sets can be disjoint.

7 The remaining real comparison

The exact target remains

$$(\forall B)(\exists m > B) \ m \nmid Z_m. \quad (17)$$

Two sufficient approaches now have explicit, different missing estimates. For the compressed primitive grids, the moments already absorb every fixed denominator. With positive moment M and upper support N , write

$$A_N = \sum_{d=D+1}^N \frac{V_d(\lambda)}{d! - 1}, \quad \mathcal{R}(\lambda) = A_N + M \sum_{d>N} \frac{1}{d! - 1}.$$

The sufficient strict comparison is

$$\frac{2M}{(N+1)! - 1} < \lfloor A_N \rfloor + 1 - A_N. \quad (18)$$

It puts the residual below the next integer and above A_N . Proving this on an unbounded grid family would suffice. The finite signed block A_N is essential. The second approach asks for the simultaneous collision and complementary-gap estimate (16). Neither estimate is established cofinally here.

The limit of short supports. The shortest-support kernels cannot meet (18). For $N = D + O(1)$, the compulsory divisibility $L_D \mid M$ and (13) give

$$\frac{2|M|}{(N+1)!-1} \rightarrow \infty,$$

whereas the right side of (18) is at most one. A useful family must therefore allow a larger upper support and still control its finite signed block.

Large denominator valuations alone also leave the real gap uncontrolled: the model fractions $1/q^e$ tend to zero. The exact lower-endpoint and doubled-prime tests require joint numerator or predecessor information. Their thresholds, the adjacent-window collapse, fixed-owner absorption, and the strongest countermodels are retained in the long reasoning record.

8 The finite evidence

The exact GMP carry certificate covers $3 \leq m \leq 300000$. Its unit carries occur at

$$52, 591, 1030, 1407, 1438, 2164, 4258, 10991, 21236.$$

In particular $b_{300000} \neq 1$, yielding $q \nmid 299999!$ by (1). This excludes every divisor of that factorial, including large divisors. It does not exclude all integers with small prime factors, since their multiplicities can be too large.

The separate 80000-bit exact rational enclosure forces 23449 common continued-fraction partial quotients. Its best-approximation certificate gives $q \geq 2^{39990} > 10^{12038}$. The open-versus-closed complete-quotient endpoint convention matters for a sharper extraction; no stronger numerical floor is asserted here. The exact sources, enclosure and integer-computation receipts are retained with the long record. Neither finite calculation supplies (17).

Statements and declarations

The finite divisor-event identities, isolated-channel recursion, vanishing moment, individual channel values, and the factor 12 are checked in `DivisorChannelBasis.lean`. The unique integral basis expansion, attainable-moment formula and compressed primitive-grid construction are ordinary proofs, described in `DivisorChannelBasis.md` and, for the grid construction, in `CompressedPrimitiveChannelKernel.md`. The quadratic stopping theorem is checked in `PaperCompleteMomentHorizon.lean`, together with the scalar recurrence, the equal-block divisibility and the prime anchor it uses. The earlier lcm-envelope certificate and depth-4 moment 1380 are ordinary proofs with exact checks in `check_moment_saturation.py`. The finite-source checks are in `check_finite_source.py`. The inherited generic lemmas in `TailIdealCertificate.lean` assume the lcm envelope; they do not formalise the factorial arithmetic. The growth $\liminf$ is the ordinary asymptotic consequence of the finite terminal-block inequality; its eventual form, with the constant $2\sqrt{2}/3$, is checked in `PaperCompleteAsymptotics.lean`.

The exact companion and carry criteria are represented by `CompanionOrbitRationality.lean` and `FactorialZeroPlateau.lean`; the unconditional correction (4) is in `CompanionConstantCarryLaw.lean`. The pole-residue identity and the two displayed reciprocal

equalities are in `PrimePoleCriterion.lean`. The complete maximal-hit data come from the separate exact modular scan. The carry census and continued-fraction certificate use exact integer computation outside Lean.

The inherited public source pin is `f36a98bf3d3e`. Source-current additions and that public snapshot must be reconciled at one immutable release before its links are represented as a replay of the whole note. Formal proof checking does not assess external novelty or the significance of the remaining hypothesis. The complete source register, receipts and failed-route calculations belong to the long record.

References

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